

System Administration & Maintenance

Network Administration Fundamentals

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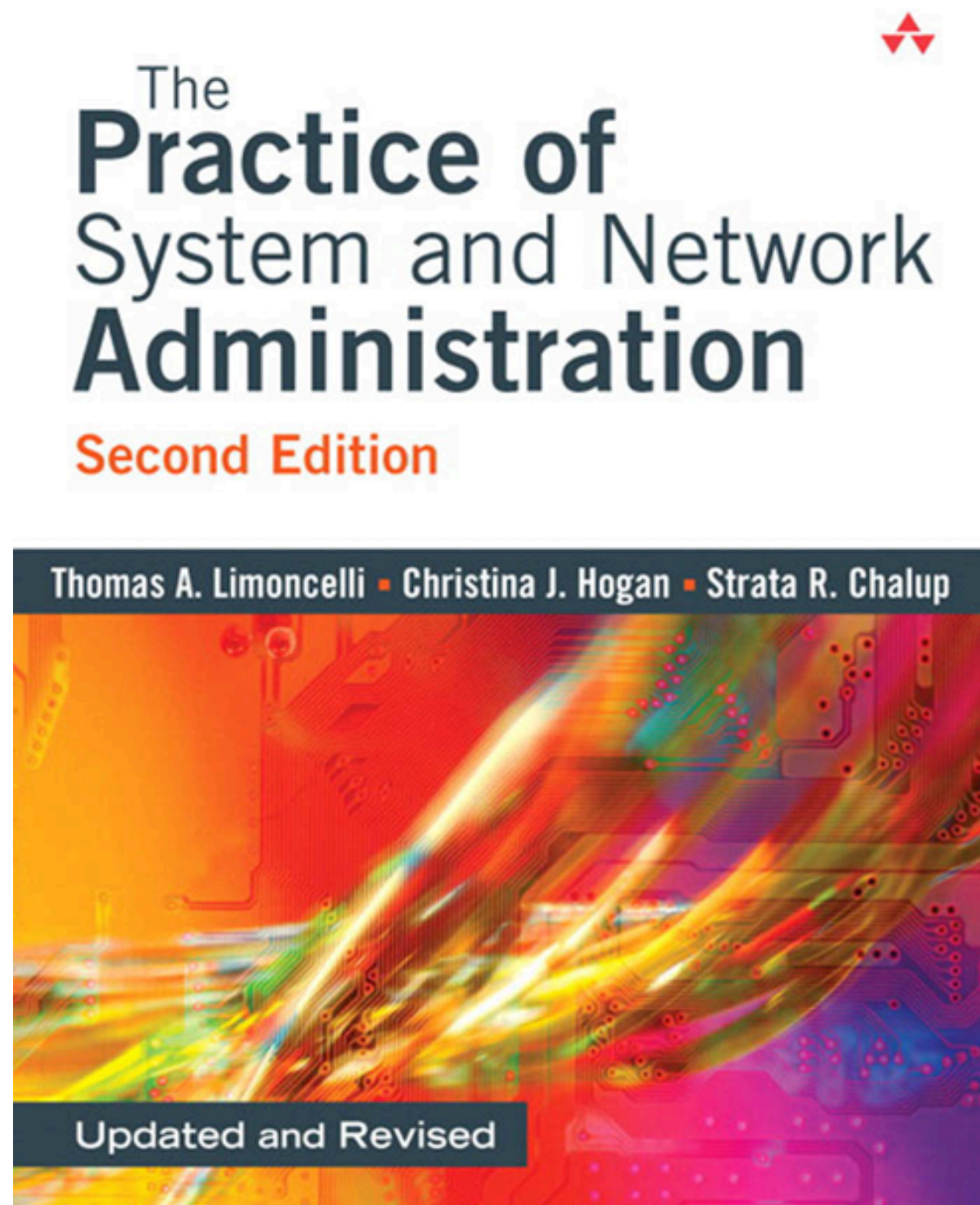
Learning Outcomes

By the end of this lecture, students should be able to:

- Explain the role of networks in system administration
- Identify common network devices
- Understand IP addressing, DNS, DHCP, and routing at a high level
- Explain the responsibilities of a network administration

References You Can Look At!

1.



What is a Network?

A network is a collection of devices connected together to exchange information.

Ex.

- Office network
- University network
- Internet
-

Why Networks are Important?

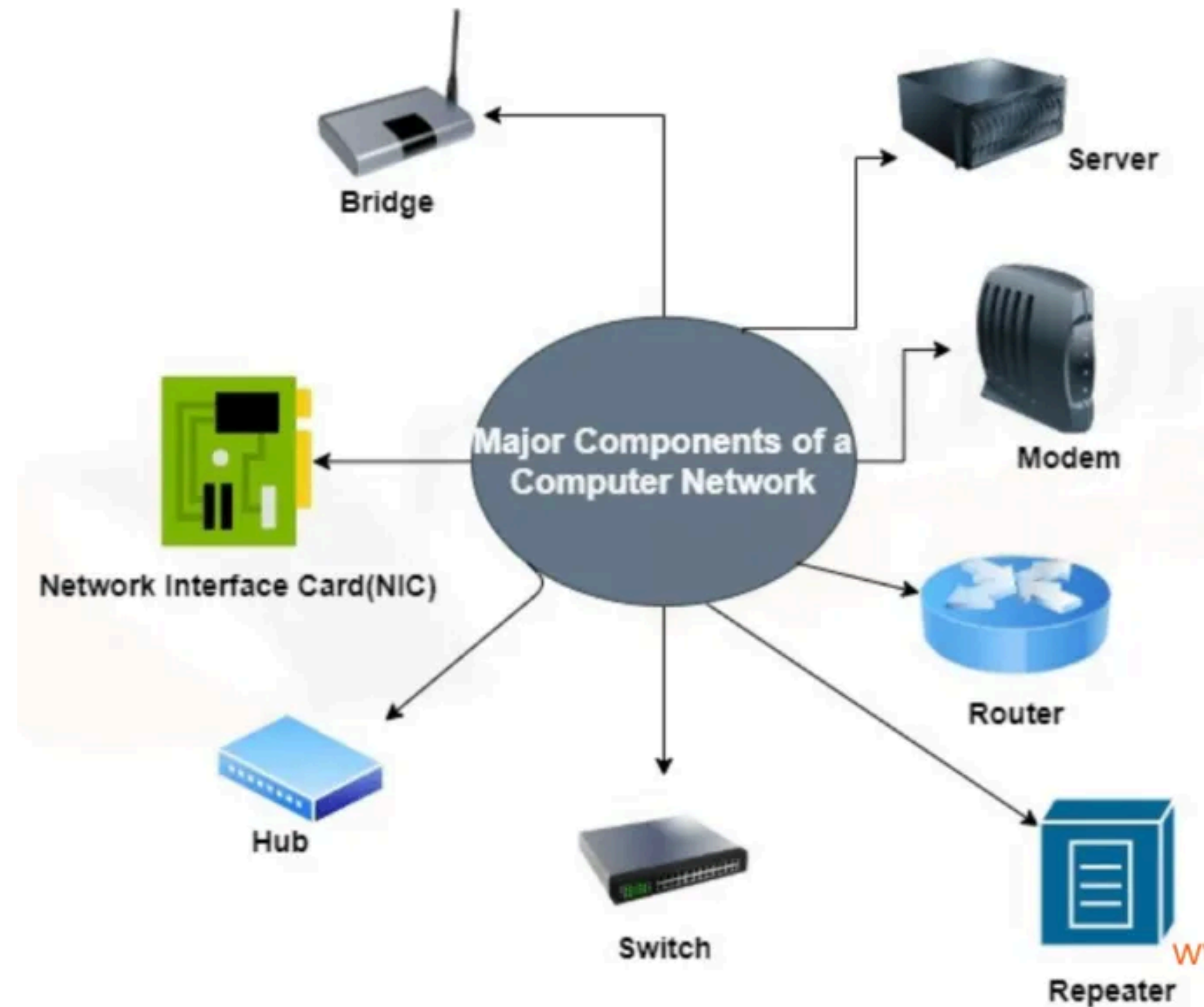
Modern services depend on networks:

- Email
- File sharing
- Web servers
- Databases
- Cloud services

Because of this: **No network = no service**

Common Network Components

- Computers
- Servers
- Printers
- Switches
- Routers
- Firewalls
- Wireless Access Points



End Devices

Devices used by users or services to consume network traffic to do specific tasks.

- Desktop Computers
- Laptops
- Servers
- Printers
- Mobile Devices

Network Devices a SA Encounters

System administrators frequently work with;

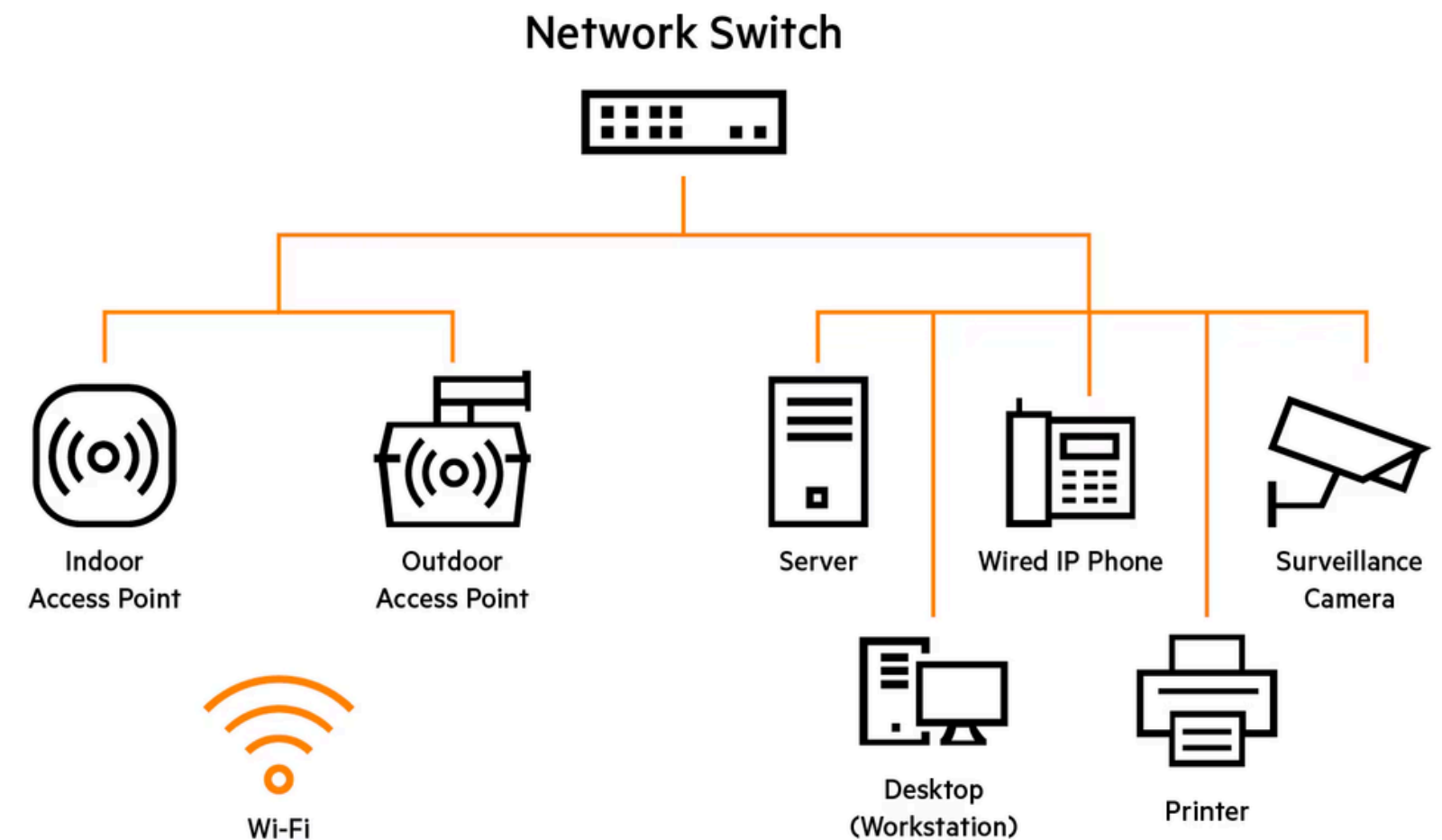
- Switches
- Routers
- Firewalls
- Wireless Access Points
- Network Attached Storage (NAS)

Network Devices a SA Encounters

- **Switch**

- A switch is a networking device used to connect multiple devices within the same Local Area Network (LAN).

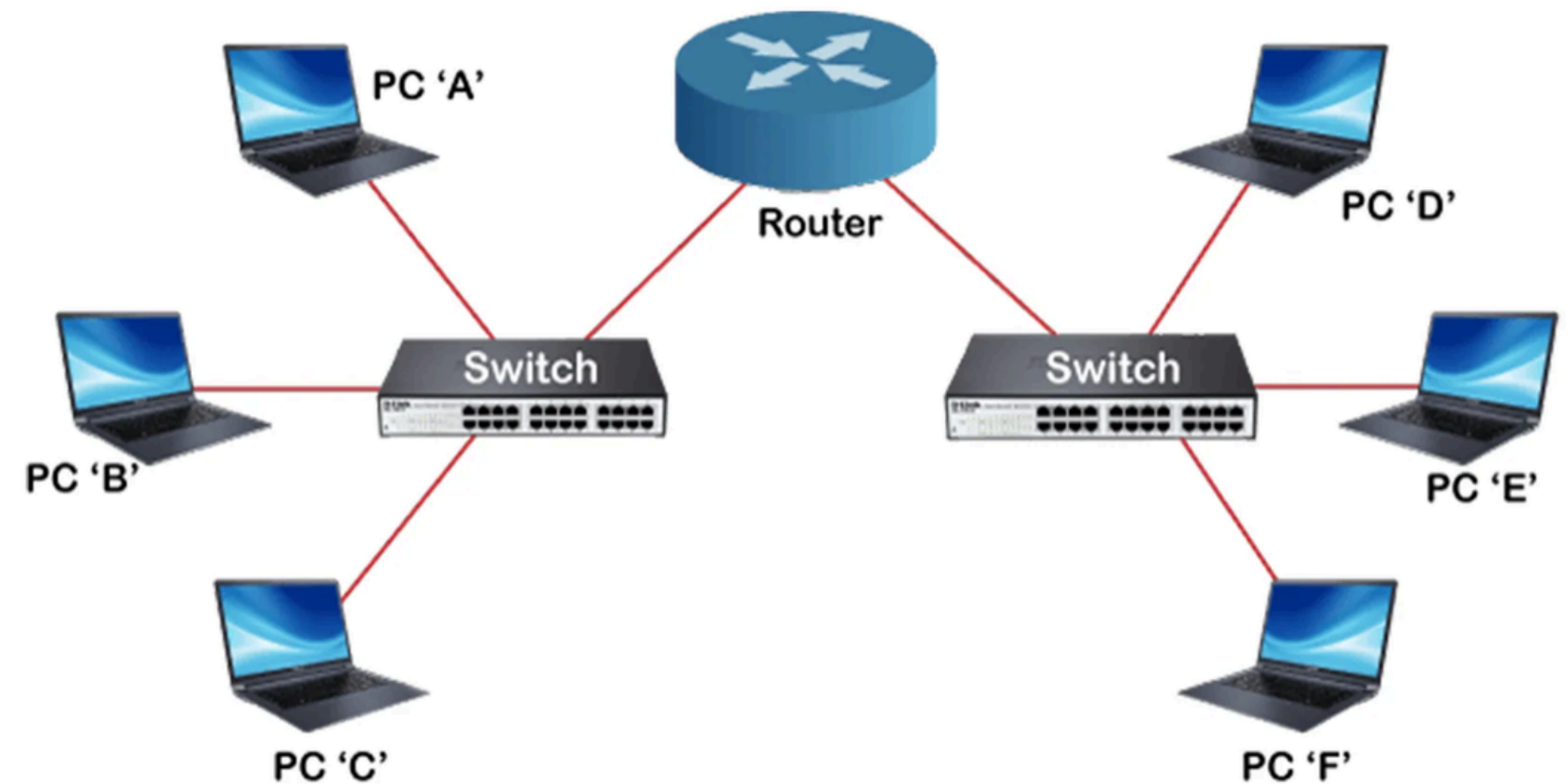
- Switches allow network administrators to isolate traffic, set policies, and segment networks.



Network Devices a SA Encounters

- **Router**

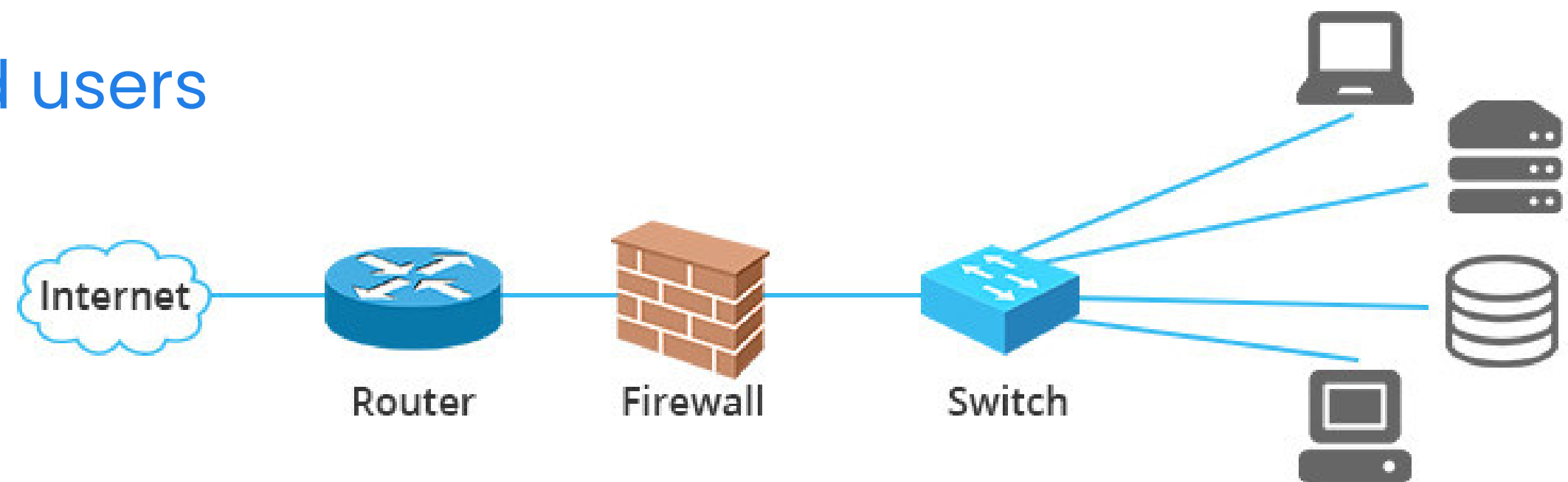
- A router connects a local network to the outside world (the Internet) or links two separate networks.
- Operates at Layer 3 (Network Layer)
- When users cannot access external resources, the router is often one of the first devices to investigate.



Network Devices a SA Encounters

- **Firewall**

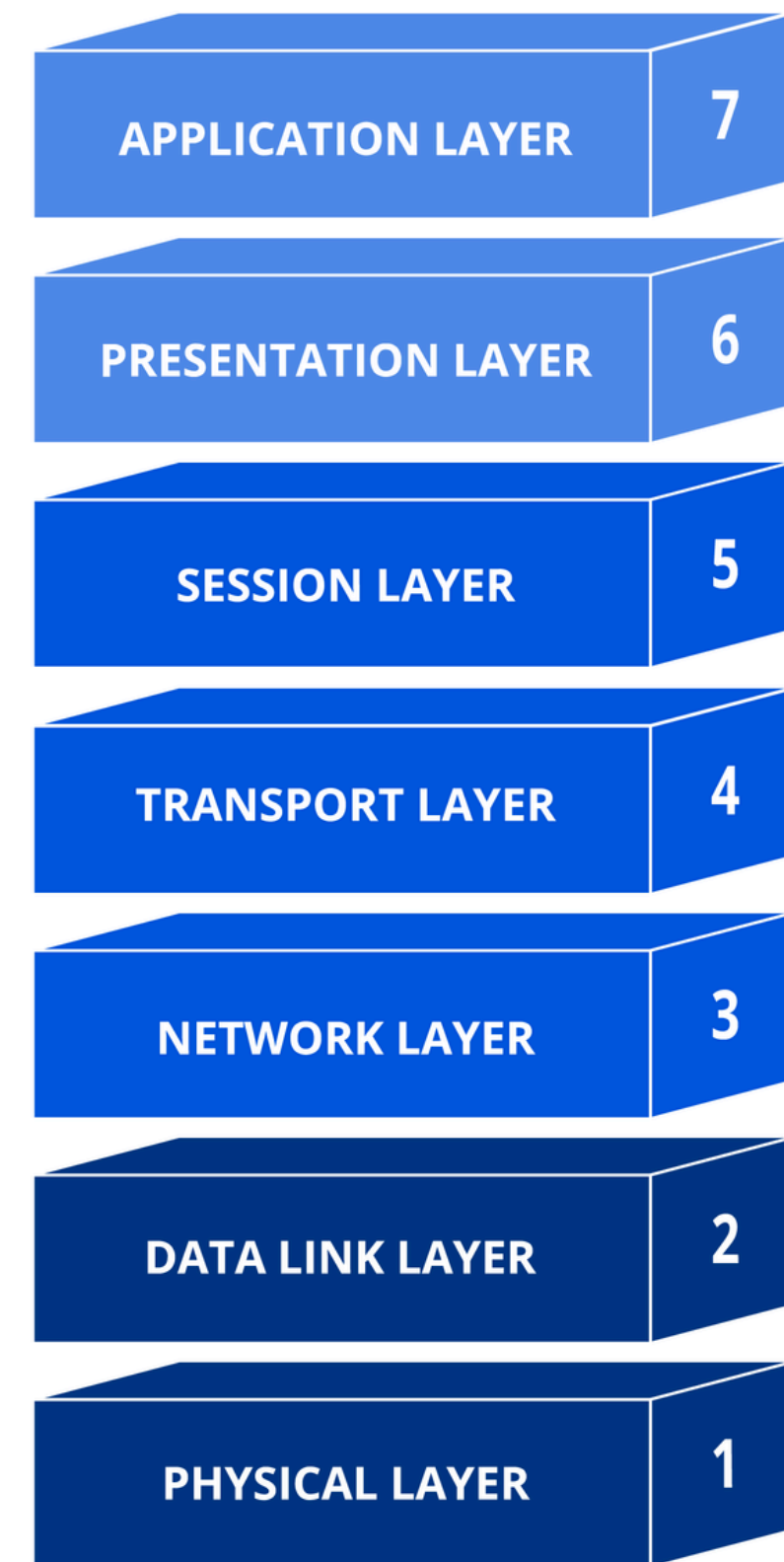
- A firewall controls network traffic entering and leaving a network.
- Prevent unauthorized access
- Enforce security policies
- Protect servers and users



OSI Model

The OSI (Open Systems Interconnection) Model is a conceptual framework that standardizes how different computer systems communicate over a network.

- The model looks at the network as 7 logical layers.



Layers of OSI Model

- **Layer 1: Physical**

- The physical and electrical hardware. It transmits raw bit streams (0s and 1s) over a communication medium.
- Examples: Cables, fiber optics, network adapters, hubs. [[1](#), [2](#), [3](#), [4](#)]

- **Layer 2: Data Link**

- **F**acilitates data transfer between devices on the same local network.
- It also handles hardware addressing (MAC addresses).

Layers of OSI Model

- **Layer 3: Network**

- Determines the best path for data to travel across a physical network (routing) and handles logical addressing (IP addresses).
- If the two devices communicating are on the same network, then the network layer is unnecessary.

- **Layer 4: Transport**

- Responsible for end-to-end communication between the two devices.
- Ensures data is delivered reliably and in the correct order. It divides data into segments and handles error-checking.

Layers of OSI Model

- **Layer 5: Session**

- This is the layer responsible for opening and closing communication between the two devices.
- The time between when the communication is opened and closed is known as the session.
- The session layer ensures that the session stays open long enough to transfer all the data being exchanged, and then promptly closes the session in order to avoid wasting resources.

Layers of OSI Model

- **Layer 6: Presentation**

- Responsible for preparing data so that the application layer can use.
- The presentation layer is responsible for the translation, encryption, and compression of data.

- **Layer 7: Application**

- This is the only layer that directly interacts with data from the user. Software applications like web browsers and email clients rely on the application layer to initiate communications
- Application protocols, e.g., SMTP (email), HTTP (web), and FTP (file transfer)

Responsibilities of Network Administration

System administrators commonly;

- Creating a clean architecture
- Monitor connectivity
- Maintain network services
- Troubleshoot issues
- Document network infrastructure
- Coordinate with network engineers

Clean Architecture

A network architecture should be as clean and simple to understand as it can be.

- It should be possible to briefly describe the approach used in designing the network.
- A clean architecture makes debugging network problems much easier.
- A clean architecture encompasses both physical and network protocols that are used on both hosts and network equipment.

Network Monitoring

Monitoring helps System administrators detect problems before users report them.

- Server unreachable
- High bandwidth usage
- Service outages
- Network failures

Keeping a Good Documentation!

Network documentation is the practice of recording and storing essential information about a network's infrastructure.

- **Network diagrams:** Visual representations of the network layout, showing physical and logical connections.
- **Device configuration details:** Documentation of device settings, firmware versions, IP addresses, and passwords.
- **Inventory management:** Keeping track of all devices, software, and other network assets.

Keeping a Good Documentation!

- **Security policies and access controls:** Defining who has access to specific areas of the network and how security is managed.
- **Incident logs and troubleshooting records:** Documenting past incidents and their resolutions for future reference.

Why Is Network Documentation Important?

- **Enhanced Troubleshooting Capabilities**
 - Faster identification of issues.
 - Clearer troubleshooting paths, minimizing downtime.
 - Reduced reliance on tribal knowledge, making the process more efficient

Why Is Network Documentation Important?

- **Facilitates Network Scaling and Upgrades**
 - Reduces the complexity of integrating new devices or expanding networks.
 - Minimizes the risk of errors or conflicts during upgrades.
 - Ensures seamless expansion without service interruption.

Why Is Network Documentation Important?

- **Supports Disaster Recovery and Business Continuity**
 - Faster recovery times minimize operational disruptions.
 - Greater clarity in emergencies and reducing stress on IT teams.
 - Allows for proactive disaster planning, ensuring continuity in critical operations.

Why Is Network Documentation Important?

- **Strengthens Network Security.**
 - Enhances the ability to perform security audits and identify vulnerabilities.
 - Supports compliance with regulations such as GDPR, HIPAA, or PCI-DSS.
 - Reduces the risk of unauthorized access or misconfigurations.

Why Is Network Documentation Important?

- **Simplifies Onboarding and Collaboration**
 - New hires or third-party contractors can get up to speed much faster.
 - Instead of exploring the network blindly, they have direct access to services, server purposes, and base standards.